

An exact null for the Rasch change index

Why the RCI critical value is not 1.96, and how to compute it without simulating

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2026-09-12

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Purpose

`RMpersonChange()` ships `critical = "simulate"` rather than the conventional 1.96, because the Rasch change index

$$RCI = \frac{\hat{\theta}_2 - \hat{\theta}_1}{\sqrt{SE(\hat{\theta}_1)^2 + SE(\hat{\theta}_2)^2}}$$

is not standard normal under the null when the scale is short. This note shows that the null does not need simulating at all: for complete data it can be enumerated exactly. It also records what the exact critical value does across test length, sample skew and targeting, and one consequence for the RCI as a statistic.

Why the null is not $N(0, 1)$

Two mechanisms, both weakening as test information grows.

Self-normalisation. $SE(\hat{\theta}_1)$ and $SE(\hat{\theta}_2)$ are deterministic functions of the realised scores, so the denominator is not independent of the numerator. A respondent who moves to an extreme score produces a large numerator and a large denominator together, and the ratio is damped. Both tails of the null are pulled in, so the critical value sits **below** 1.96. In a t-test the denominator is independent of the numerator; here it is not, which is why the t reference does not help either.

Discreteness. With complete data the sum score is sufficient, so $\hat{\theta}$ takes only as many values as there are scores and the null is discrete rather than continuous.

Exact enumeration

Sufficiency is what makes this tractable. $\hat{\theta}(r)$ and $SE(r)$ are deterministic lookups, the ones `RMscoreSE()` builds, and $P(r | \theta)$ follows from the Lord-Wingersky recursion. Under the null the two occasions are independent given θ , so the joint is $P(r_1 | \theta)P(r_2 | \theta)$ and the whole null is a discrete distribution over $(k + 1)^2$ score pairs. For a six-item partial credit model that is 361 cells.

```
# Lord-Wingersky: exact sum-score distribution given theta under a PCM
score_dist <- function(thr_list, theta) {
  d <- 1
  for (thr in thr_list) {
    p <- easyRasch2:::pcm_cat_probs(theta, thr)
    nd <- numeric(length(d) + length(p) - 1L)
    for (a in seq_along(d)) {
      nd[a + seq_along(p) - 1L] <- nd[a + seq_along(p) - 1L] + d[a] * p
    }
    d <- nd
  }
  d
  # index is score + 1
}
```

```

}

# Exact null distribution of the RCI, pooled over a set of person locations
exact_crit <- function(thr_list, thetas, alpha = 0.05) {
  steps <- vapply(thr_list, length, integer(1L))
  lut <- vapply(0:sum(steps), function(r) {
    easyRasch2:::.theta_wle(
      easyRasch2:::.score_pattern(r, steps), thr_list, c(-10, 10)
    )
  }, numeric(2L))
  th <- lut[1L, ]
  se <- lut[2L, ]
  rci <- outer(seq_along(th), seq_along(th),
    function(i, j) (th[j] - th[i]) / sqrt(se[i]^2 + se[j]^2))
  w <- matrix(0, length(th), length(th))
  for (t in thetas) {
    d <- score_dist(thr_list, t)
    w <- w + outer(d, d)
  }
  w <- w / sum(w)
  o <- order(as.numeric(rci))
  v <- as.numeric(rci)[o]
  cw <- cumsum(as.numeric(w)[o])
  c(lower = v[which(cw >= alpha / 2)][1L],
    upper = v[which(cw >= 1 - alpha / 2)[1L]])
}

```

Validation against simulation

Run by `dev/` scratch scripts at `sim_iter` 2000 to 4000, $n = 200$, $\alpha = .05$ two-sided. The simulated and exact values agree to within Monte Carlo error in every configuration tried, including dichotomous items, items with unequal category counts, a right-skewed person distribution and a badly off-target one.

configuration	simulated	exact	difference
6 polytomous items, symmetric theta	-1.695 / +1.695	± 1.695	0.000
6 polytomous items, skewed theta	-1.725 / +1.695	± 1.695	0.030
6 polytomous items, skewed and off target	-1.534 / +1.534	± 1.522	0.012
20 dichotomous items, symmetric	-1.798 / +1.798	± 1.798	0.000
20 dichotomous items, skewed	-1.798 / +1.798	± 1.798	0.000
mixed 2/3/4 categories, symmetric	-1.721 / +1.721	± 1.721	0.000
mixed 2/3/4 categories, skewed	-1.721 / +1.721	± 1.721	0.000
12 polytomous items, symmetric	-1.864 / +1.864	± 1.862	0.002

The exact values are perfectly symmetric. That is the expected result: under the null the two occasions are exchangeable, so $\hat{\theta}_2 - \hat{\theta}_1$ is symmetric about zero however skewed $\hat{\theta}$ itself is. The residual asymmetry in the simulated column is Monte Carlo noise. Asymmetry becomes real only when exchangeability fails, which in practice means the two occasions differing in which items were answered.

The critical value tracks test information

```
set.seed(1)
theta <- rnorm(400, 0, 1.4)
tab <- t(vapply(c(4, 6, 10, 20, 40), function(k) {
  thr <- lapply(seq(-1.2, 1.2, length.out = k), function(b) b + c(-0.8, 0, 0.8))
  c(items = k, exact_crit(thr, theta)[["upper"]])
}, numeric(2L)))
kable(tab, digits = 3, col.names = c("Items", "Critical value"))
```

Items	Critical value
4	1.619
6	1.695
10	1.797
20	1.854
40	1.918

Monotone in test length, converging on 1.96. Referring the RCI to a normal null is therefore conservative, and increasingly so the shorter the scale. On a six-item scale, using 1.96 instead of 1.695 is a materially stricter test.

Skew and targeting

Sample skew on its own barely moves the pooled critical value: the skewed and symmetric rows above agree to the third decimal in five of seven configurations. Targeting is what matters. Pushing the same skewed distribution 2.2 logits off target, so that 39.5% of respondents sit at an extreme score at one occasion or the other, drops the critical value from 1.695 to 1.522. The mechanism is the self-normalisation above: a floor-bound sample spends more of its time where the statistic is most heavily damped.

This also means the pooled critical value is a property of the sample as well as the items, which is worth stating wherever it is reported.

A consequence for the RCI as a statistic

Because the denominator is a function of the same scores as the numerator, the RCI does not order respondents by how much they changed. A concrete pair from the six-item scale above:

	score change	change in logits	RCI
A	0 → 18	7.24	3.53
B	3 → 15	3.11	3.60

Respondent A moved the full width of the scale, 4.14 logits further than B, and scores lower on the RCI. Both are far past any critical value so nothing turns on it here, but it shows that the RCI is a standardised distance from the null rather than a measure of change, and should not be read as the latter.

The constructive version: since the critical value has to be computed from the item parameters anyway, the normalisation buys nothing that a location-specific threshold on the raw change would not. The same enumeration gives the null of $\hat{\theta}_2 - \hat{\theta}_1$ directly, so a conditional critical **change**, in logits, is available at no extra cost and is monotone in what happened. That is the same object as a conditional minimal detectable change, and connects to the score-level MDC table parked as a possible sibling function.

Scope and limits

- **Complete data only.** Missing data breaks sufficiency of a single sum score. Each distinct missingness pattern has its own score distribution and its own lookup, so the enumeration still works but per pattern.
- **null = "retest" needs quadrature.** The occasion deviations are continuous, so the joint becomes a double integral over u_1, u_2 rather than a pure sum. Still deterministic and cheap at, say, 21 nodes a side.
- **EAP is fine.** On complete data an EAP estimate is also a function of the sum score.
- Validated here for WLE with CML thresholds. Not yet checked against MML item parameters.

Implemented

Shipped as `critical = "exact"`, now the default in `RMpersonChange()`, in `R/person_change_exact.R`. Measured against a 4000-iteration simulation on the same data: critical values agree to within Monte Carlo error, p-values correlate at 0.9998, classifications are identical, and the exact path takes 0.08 s against 13.7 s. Incomplete data is enumerated within each missingness pattern, `null = "retest"` integrates the occasion deviations out by quadrature (they are independent, so the joint factorises and the retest null costs

no more than the measurement null), and EAP is supported by holding the prior at the one the observed respondents were scored under.

Having an exact reference immediately paid for itself: it exposed a bug in the simulation path, which had been re-estimating the EAP prior from each null dataset rather than holding it at the observed one. That is now fixed, and the two paths agree to four decimals under EAP.

It also removes the one real objection to `conditional_crit = TRUE`: a per-respondent critical value from simulation rests on `sim_iter` draws, roughly 25 of which land in each tail at $\alpha = .05$, whereas $P(r \mid \theta_i)$ is available per respondent for free.

It would also sharpen the planned simulation study, whose first aim is the realised Type I error of the simulated critical value. Much of the noise in that comparison disappears if the reference is exact.

Reproduction

Scratch scripts used for the simulated column are not part of the package. The exact computation is self-contained in the `exact-fns` chunk above and needs only `easyRasch2:::.pcm_cat_probs()`, `:::.theta_wle()` and `:::.score_pattern()`.